Comparing Party Systems

A Multidimensional Approach

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The comparative study of political parties once consisted largely of attempts to differentiate among one-, two-, and multi-party systems. Eventually, however, the extreme oversimplification of this classification scheme came to be recognized, and more elaborate schemes which contained categories like "modified one-party system" and "two-and-a-half party system" began to appear. Unfortunately, these new classifications proved to be extremely arbitrary in the criteria that were used to classify party systems and highly capricious in the way that criteria were applied.[1] Dissatisfied with the classificatory approach, comparativists then began to develop numerical indices which they hoped would facilitate systematic analysis. Over the last fifteen years, literally dozens of these indices have made their way into the research literature.[2] The purpose of each such index is to assign a single numerical value to a party system for use in comparing intersystem differences at a point in time, intrasystem differences over time, or a combination of both.

In sharp contrast to both the traditional classificatory approach and the more recent index-construction approach, we begin with the assumption that party systems are inherently *multidimensional*. While the indices which have flourished in recent years are useful for certain purposes, they—like the earlier classification schemes—greatly understate the complexity and diversity of the party systems that one finds worldwide. The main virtue of existing indices is that they focus attention on a single aspect of the operation of party systems and treat that aspect in an "objective," replicable fashion. However, this is also their primary defect, for in focusing attention on only one aspect of party systems they forfeit the ability to capture other important aspects as well. In our view, a multidimensional approach, in which several different aspects of party systems simultaneously serve as the bases of comparison, is needed to sensitize comparativists to the considerable differences that sometimes characterize systems which seem very similar when viewed unidimensionally, as well as to the unexpected similarities that sometimes crop up between systems which are generally thought to be very different.

This paper begins with a discussion of four attributes that we consider central to the analysis of party systems. While these attributes have been utilized in previous research on party systems, each is normally analyzed in isolation from the others. We, on the other hand, discuss a procedure that facilitates comparison based upon the simultaneous consideration of all four attributes. Utilizing this procedure, we employ data from forty-six nations to develop a multidimensional representation of party systems. We then discuss the potential theoretical importance of the information gained from this multidimensional representation.

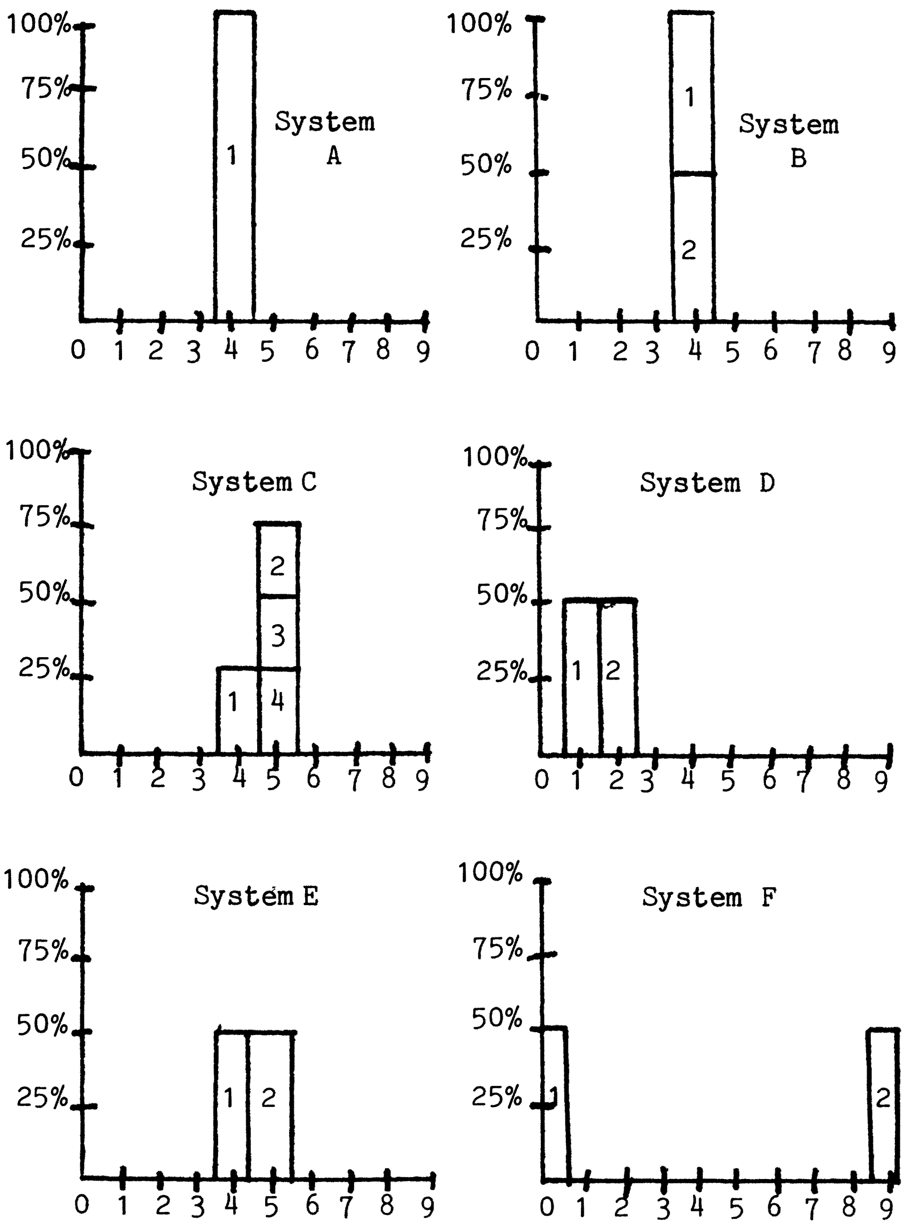
The Dimensions of Comparison

Underlying virtually every classification system and numerical index that have been devised for the comparison of party systems is the idea, whether explicit or implicit, that party systems are structures of representation. We have no quarrel with this idea, for a primary function of political parties is to offer candidates for political office, and one of the most obvious bases of party success is the frequency with which the candidates who campaign under its banner are elected. What we find inappropriate is the idea that party systems are *exclusively* structures of representation. Regardless of how successful they may be in winning elections, parties also exist to aggregate and articulate certain political points of view. In this sense, party systems can also be looked upon as structures of articulation. Comparison might focus, then, not only upon the competition among various party organizations, but also upon the competition among various ideological-programmatic outlooks. We focus upon the ideological aspects of articulation because ideologies are, to an extent unrivaled by other potential aspects of articulation (such as group interests), comparable across systems.

Taken together, these two aspects of the operation of party systems allow us to construct party systems profiles in a fashion very similar to a city skyline. The vertical dimension of a system profile relates to each party's degree of success in terms of the representation dimension, the operational measure of which is each party's percentage share of seats in the lower house of the national legislature.[3] This aspect of the party system profile has attracted considerable attention in the recent past, as the proliferation of indices relating to fractionalization, fragmentation, competition, and a host of related constructs attests. The horizontal dimension, on the other hand, relates to the ideological-programmatic position of each party, irrespective of its electoral successes or failures. Distance in this frame of reference is ordinarily measured in terms of parties' positions on the classic left-right issue, that of government intervention in the economy. We realize that there are systems in which the basic lines of political cleavage may be largely unrelated to the left-right dimension, adhering instead to religious, racial, ethnic, linguistic, regional, or personalistic lines. Still, previous comparative work in parties and party systems has established that parties in widely divergent cultural contexts can meaningfully be compared with respect to their standing on the left-right dimension,[4] and the spatial view of the operation of party systems[5] continues to exercise great sway over theoretical work on parties and party systems. In what follows, then, we concentrate on the left-right aspect of differences among parties because this aspect is both theoretically important and amenable to cross-national comparison; in those relatively infrequent instances where the parties in a particular system do not lend themselves to this type of treatment, we shall simply drop the system from consideration.[6]

The skyline metaphor becomes clearer when we turn to Figure 1, which presents a set of hypothetical party system profiles. Assume for the moment that a party can be characterized in terms of its position on a numerical left-right scale; it will be convenient to think in terms of a ten-point scale. Party system profiles can then be

Figure 1 Ideological Distributions of Six Hypothetical Party Systems



generated simply by plotting each party's left-right score on the horizontal axis, against its share of seats in the lower legislative chamber on the vertical axis. In histogram form, the two dimensions show the manner in which parties at various points on the ideological continuum contest political offices—considerably more information than would be provided if only one of the two dimensions was considered.

A great deal can be learned from simple visual inspection of party system profiles like those contained in Figure 1, but the systematic analysis of party systems can hardly end with such visual inspection. Analytically, the problem becomes one of summarizing the information in a party system profile in a manner that facilitates comparison. This goal can be achieved through the computation of four separate summary statistics for each party system.

The first and in some ways the most basic thing to know about a party system pertains solely to the vertical dimension: Does one party dominate the system, or is the system more competitive? The most widely used and, for most purposes, the most satisfactory way to answer this question is to calculate a fractionalization index summarizing the distribution of seats among the parties in a system, according to the following formula.[7]

$$\text{Party fractionalization} = 1 - \sum_{i=1}^{n} T_i^{\,2},$$

where T_i is a party's decimal share of seats. To illustrate, in Figure 1 one party entirely dominates System A, making it a perfectly nonfractionalized system (party fractionalization = 0). In both System B and System C, seats are divided equally among all the parties in the system. System C has four parties, each with 25 percent of the seats, while System B has only two parties, each with 50 percent of the seats. Thus C is more highly fractionalized (party fractionalization = .75 for C and .50 for B).

Any two parties in a particular party system may or may not occupy the same position on the left-right continuum. Accordingly, knowing how fractionalized a party system is in terms of the distribution of legislative seats among various parties will not necessarily tell us how fractionalized it is in terms of the distribution of seats among various ideological-programmatic positions.[8] For example, the two parties in System B are ideologically indistinguishable, so System B is no more ideologically fractionalized than System A is (ideological fractionalization = 0). To take a different example, in System D there are two ideologically distinct parties which command 50 percent of the legislative seats apiece. By contrast, in System C Party 1, with 25 percent of the seats, is ideologically distinctive, but Parties 2, 3, and 4, each with 25 percent of the seats, are ideologically indistinguishable from one another; thus, System C, which scores higher than System D on the party fractionalization index, turns out to be less fractionalized than D in terms of ideology (ideological fractionalization = .375 in C and .50 in D). More generally,

$$\text{Ideological fractionalization} = 1 - \sum_{i=1}^{n} I_i^{\,2},$$

where I_i is the decimal share of seats held by parties at each ideological position on the left-right scale.

These two indices, however, do not exhaust the range of information that is needed to compare the party system profiles. Consider now Party System E, which is identical to System D with respect to both party and ideological fractionalization; in each system, two ideologically distinctive parties split the seats evenly. Yet, Systems D and E are not identical, for the two parties are both to the left of center in System D but centrist in System E. What varies from System D to System E, then, is the ideological center of gravity of each, and failure to take account of this difference could lead one to miss a probable cause of variation between the systems, for example in the policy outputs they produce. Statistically, measurement of a system's center of gravity is straightforward; one simply calculates a weighted mean.

$$\text{Ideological center of gravity} = \sum_{i=1}^{n} T_i C_i,$$

where T_i is a party's decimal share of seats, and C_i refers to the party's position on the ideological continuum, expressed numerically. If, for example, Parties 1 and 2 in System D were assigned scores on the ten-point ideological continuum of 1 and 2, respectively, then the ideological center of gravity for System D would be 1.5; by the same token, if Parties 1 and 2 in System E were coded 4 and 5 on the ideological scale, then the ideological center of gravity for System E would be 4.5.

The need for one final summary statistic becomes evident when we compare Party Systems E and F. These two systems are identical in terms of party fractionalization (.50), ideological fractionalization (.50), and ideological center of gravity (4.5). However, the two systems are still very different from one another, for System E has a decided centrist orientation while System F is highly polarized. From a Downsian perspective, the potential importance of this distinction is clear: the likelihood for long-term stability is greater for System E than for System F.[9] In order to statistically summarize differences such as that between Systems E and F, a measure of deviation within party systems is needed. For this purpose, we use the weighted mean difference as a measure of ideological polarization.

$$\text{Ideological polarization} = \sum_{j=1}^{n} \frac{\sum_{K=1}^{n} t_K \left| x_j - x_K \right|}{1 - t_j}$$

where t_j equals party j's decimal share of the seats; t_k equals party K's decimal share of the seats; X_j equals part j's position on the left-right continuum; X_K equals party K's position on the left-right continuum; and n equals the number of parties in the system. The discriminatory power of the ideological polarization dimension can be seen by comparing Systems E and F. While their scores on the other indices are identical, System E has an ideological polarization score of 1 while System F has a score of 9.

Every party system can be seen as occupying a position on the continuum representing each attribute. Moreover, the attributes are, with the exception mentioned in note eight above, separate from one another. For example, a system can be highly ideologically fractionalized but only minimally ideologically polarized, or vice-versa. Empirically, of course, any two attributes may prove to be interrelated. What remains unspecified at this point is how to compare systems based upon a simultaneous consideration of all four attributes. In the next section, we consider a multidimensional procedure that facilitates such comparison.

Multiattribute Comparisons among Party Systems

Let us begin with the simple case of characterizing party systems in terms of the party fractionalization index. The range of the index is 0 to 1, with each party system being assigned a value somewhere in this range. If we want to know how similar two party systems are in terms of this attribute, we merely take the absolute value of the difference between the scores associated with each. The dimensional analogue of this process is to conceive of the index in terms of a line segment with each party system represented by a point on the line. The similarity of two party systems is now represented by the distance between the two points representing the two party systems.

The simplicity of the unidimensional case can easily be extended to the multidimensional case. It is this multidimensional case that allows us to compare party systems in terms of all four attributes. Since we are dealing with four conceptually independent attributes, each party system is represented as a point in a four-dimensional space. Thus, each party system is totally specified by a four-tuplet with the similarity between any two party systems given by the distance between the two points representing the systems in four-dimensional space.

Only one technical difficulty must be solved before we are ready to proceed: We need to assign numerical quantities to the interpoint distances. The classical formula for computing interpoint distances in a four-dimensional space is given in Equation 1.

$$(1)\ d^2_{ij} = \sum_{t=1}^{4} (X_{it} - X_{jt})^2$$

where X_{it} equals the position of party i on dimension t; and X_{jt} equals the position of party j on dimension t. However, we cannot directly use the formula in Equation 1 because we must insure that no single dimension dominates the distance measures merely because of its defined scales. Each formula that we use to define an attribute places a different upper and lower bound on the values that can be associated with any party system. For example, the party fractionalization index has a range of 0 to 1 while the center of gravity has a range of 0 to 9. If Equation 1 is used to compute interpoint distances, other factors remaining equal, the second index will dominate the two-dimensional measures. Since we consider each attribute equally important in classifying party systems, it is necessary to compensate for the differences in scale among the four attributes.

Fortunately, traditional procedures in multidimensional scaling allow us to cope with this problem. Thus, the formula in Equation 2 is used to compute interpoint distances.

$$(2)\quad d_{ij}^2 = \sum_{t=1}^{4} (W_t X_{it} - W_t X_{jt})^2$$

where X_{it} equals the position of party i on dimension t; X_{jt} equals the position of party j on dimension t; and W_t is a linear transformation unique to each dimension. W_t is defined by Equation 3.

$$(3)\quad W_t^2 = \frac{100}{\sum_{i=1}^{n} X_{it}^2}$$

where X_{it} equals the position of party i on dimension t and n is equal to the number of party systems. The impact of utilizing Equation 3 is to normalize each dimension such that the sum of the squared distances from 0 on each dimension equals 100. Also, each dimension has the same mean of the squared values. As such, the use of Equation 2 will guarantee that no single dimension dominates the distance measures merely because of its original scale. Since the use of the W_t's results in a linear transformation of the original scale values, the original information provided by each quantitative scale is not fundamentally changed.

It is Equations 2 and 3 which permit us to simultaneously consider all four attributes. The similarities among party systems can be seen to be in a one-to-one relationship with the interpoint distances. Systems that are very similar with respect to all four attributes will have small distances among them, while systems that are very different will have large distances among them.

Comparing Party Systems Worldwide

These similarity measures can be used in a number of different ways. At this point, however, our interest lies in what amounts to an exploratory mapping operation: we wish to use them to describe and understand the structure underlying the party systems one finds worldwide.

The data for this analysis pertain to forty-six party systems scattered throughout the world. The only criteria for inclusion in the study were (1) that a party system contain more than purely token opposition, so that we would be comparing systems which were at least minimally open and competitive,[10] and (2) that, allowing for a few minor non-left-right parties, the parties within a system be codeable in left-right terms from available data. Data on these forty-six party systems were taken from the 1979 *Britannica Book of the Year,* which listed the number of seats in the lower legislative chamber held by each party as of December 31, 1978. The *Britannica Book of the Year* also assigned labels to parties on a left-right continuum ranging from fascist to communist. Ten separate positions on the left-right spectrum were distinguished: fascist; extreme right; right; center right; center; non-Marxist left;

social democratic; socialist; extreme left; and communist. For purposes of analysis, these ten positions were assigned scores ranging from 0 to 9.[11]

The analysis began by assigning the forty-six party systems a position on each of the original four dimensions. In effect, then, each party system was seen as occupying a single point in four-dimensional space. Equation 2 was then used to compute interpoint distances among the forty-six party systems.

While these 1,035 interpoint distances could be viewed from several different perspectives, we were interested in examining the empirical structure underlying national party systems. That is, even though party systems can be classified theoretically in terms of the four constructs, is this four-dimensional structure empirically useful? If so, how is this simpler structure best understood? Since each dimension and each party system were represented in terms of a spatial analogue, multidimensional scaling was deemed the most appropriate procedure to search for the simplified empirical structure.

A multidimensional analysis of the 1,035 interpoint distances indicates that the forty-six party systems could be best represented in terms of a simple two-dimensional space. Utilizing Kruskal and Carmone's multidimensional scaling procedure, MDSCAL-5M,[12] a two-dimensional representation of the forty-six party systems resulted in a stress value of only .09 (stress formula 2).[13] Given the number of points and the low stress value, it is clear that the structure underlying empirical party systems is much less complex than that inderlying our original four-dimensional conceptualization. The question of immediate importance is, then, what is the nature of this empirical structure and how does it relate to our original four-dimensional conceptual structure?

Table 1 provides the symbolic identification for each of the party systems along with the position of each party system on each dimension. Figure 2 portrays the two-dimensional representation of the forty-six party systems. The actual location of the axes was specified by performing a varimax rotation on the original solution that we obtained from the multidimensional scaling routine.[14] Since we were centrally interested in the relationship between the two-dimensional solution and the four concepts underlying our system of classification, we inserted a vector representing each of these concepts into the two-dimensional space.[15] Each of these four vectors is also shown in Figure 1.

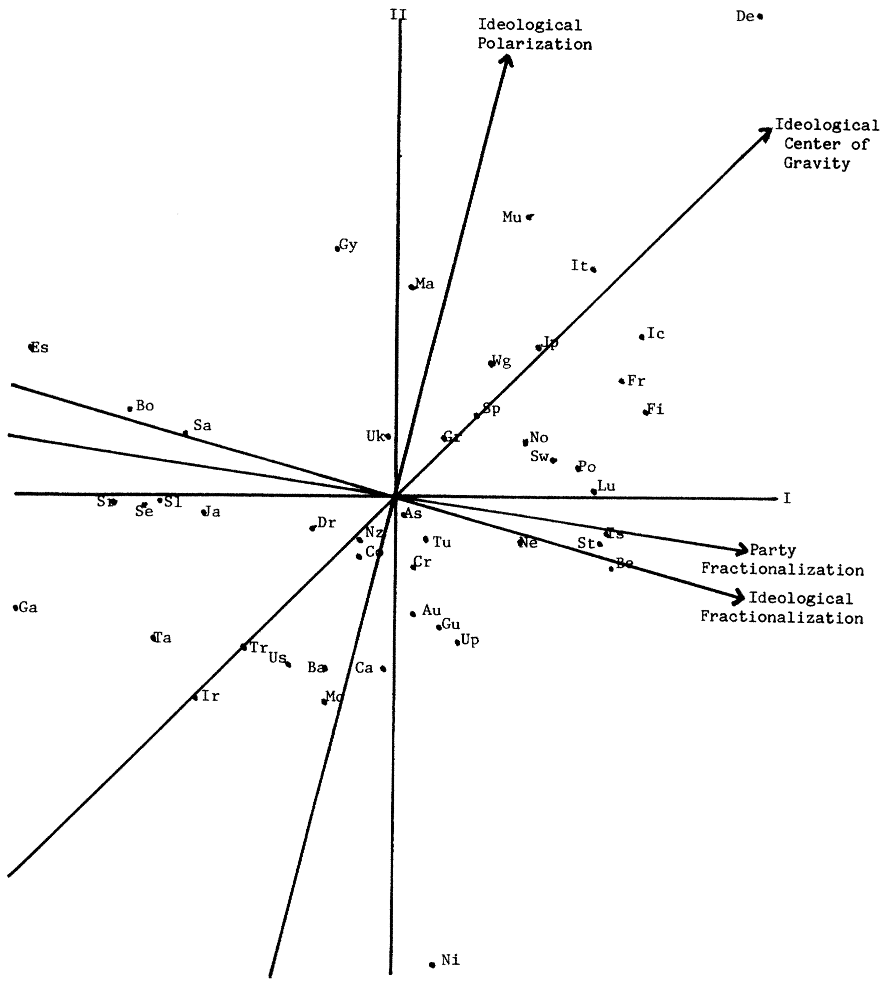
As an initial point of departure, it is important to interpret the dimensions underlying the structure in Figure 2. Dimension I is readily interpretable as a fractionalization dimension. However, there does not appear to be a significant empirical difference between ideological fractionalization and party fractionalization.[16] The party systems with the lowest scores on Dimension I are those which are least fractionalized in terms of both ideological and party considerations, while the systems with the highest scores on Dimension I are the most highly fractionalized. The vector placement of the party and ideological indices further suggests the importance of fractionalization to intersystem discrimination on Dimension I. Both vectors are closely aligned with Dimension I, with the correlations between the party systems' positions on Dimension I and their positions on the party and ideological fractionalization vectors being .94 and .94, respectively.

The second dimension discriminates among the party systems primarily on the

Table 1 Name, Symbol Identification, and Dimensional Position of Forty-Six National Party Systems

Party System	Symbol Identification	Dimensional Position Dimension I	Dimension II
Australia	Au	.07	- .49
Austria	As	.02	- .09
Barbados	Ba	- .30	- .71
Belgium	Be	.91	- .28
Botswana	Bo	-1.14	.31
Canada	Ca	- .02	- .71
Colombia	Co	- .17	- .24
Costa Rica	Cr	.05	- .28
Denmark	De	1.54	1.97
Dominican Republic	Dr	- .37	- .14
El Salvador	Es	-1.57	.61
Finland	Fi	1.02	.37
France	Fr	.93	.51
Gambia	Ga	-1.77	- .49
West Germany	Wg	.38	.57
Greece	Gr	.20	.24
Guatemala	Gu	.19	- .54
Guyana	Gy	- .31	1.02
Iceland	Ic	.97	.73
Ireland	Ir	- .85	- .85
Israel	Is	.86	- .15
Italy	It	.78	.97
Jamaica	Ja	- .84	- .07
Japan	Jp	.58	.62
Luxembourg	Lu	.84	.00
Malta	Ma	.04	.89
Mauritius	Mu	.52	1.18
Morocco	Mo	- .31	- .88
Netherlands	Ne	.52	- .18
New Zealand	Nz	- .17	- .20
Nicaragua	Ni	.15	-1.96
Norway	No	.52	.22
Portugal	Po	.75	.15
Senegal	Se	-1.04	- .06
Sierra Leone	Sl	-1.01	- .06
South Africa	Sa	- .90	.21
Spain	Sp	.34	.33
Sri Lanka	Sr	-1.19	- .03
Sweden	Sw	.68	.15
Switzerland	St	.88	- .14
Tanzania	Ta	-1.04	- .60
Trinidad	Tr	- .64	- .65
Turkey	Tu	.13	- .20
United Kingdom	Uk	- .03	.25
United States	Us	- .45	- .71
Upper Volta	Up	.24	- .60

Figure 2 Two-Dimensional Representation of Forty-Six National Party Systems and Four Party System Vectors



basis of ideological polarization. The systems with the greatest positive values on Dimension II have the highest ideological polarization scores, while the systems with the highest negative values on Dimension II have the lowest ideological polarization scores. The ideological polarization vector aligns very closely with Dimension II, with the correlation between the party systems' positions on Dimension II and their positions on the ideological polarization vector being .97.

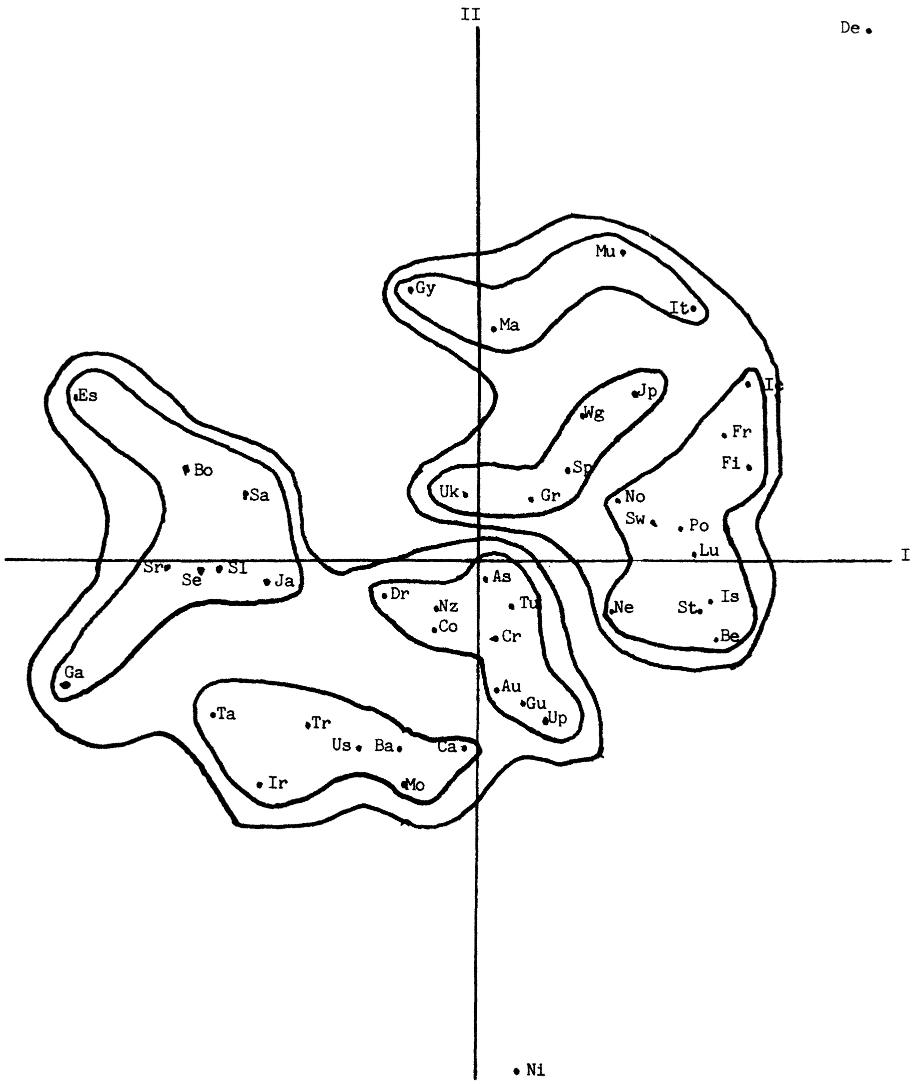
Thus, a number of interesting observations can be made about the relationship between the empirical structure and our original four concepts. First, empirically there is almost no difference between ideological fractionalization and party fractionalization. As already stated, the two constructs are highly intercorrelated, and both align quite nicely with the first dimension of our two-dimensional solution. If we reconsider the examples that were given in Figure 1, the empirical tendency is such that we do not find the simultaneous existence of Type A and Type B party systems. Only one type of system, Type A, tends to dominate the empirical world. Such a finding tends to support Downs' speculations about the relationship between ideological distributions and the number of parties in a system.[17] Turning again to the examples given in Figure 1, we saw that it is possible for one system to have a higher level of party fractionalization (System C) than another system (System D), yet have a lower level of ideological fractionalization. However, the data indicate that this does not tend to occur in the empirical world.

A second important observation is that ideological polarization appears to be critically important to the discrimination among party systems. This is reflected in the fact that Dimension II primarily discriminates among the party systems in terms of ideological polarization. In terms of the examples given in Figure 1, the difference between systems such as E and F does have empirical relevance. The data further indicate that there is a systematic relationship between ideological polarization, on the one hand, and party fractionalization, ideological fractionalization, and ideological center of gravity, on the other; the correlations are .39, .31, and −.48, respectively. Thus, systems that score higher on ideological polarization also tend to be more highly fractionalized and more leftist-oriented ideologically. Still, these correlations are not of sufficient magnitude to be exclusionary; that is, there are systems that are low in terms of ideological polarization which are among the more fractionalized and more conservative. As such, ideological polarization remains a critical component in analyzing differences among party systems.

It is particularly interesting to observe the empirical relevance of the ideological center of gravity construct. The ideological center of gravity is not sufficiently independent of the other dimensions to require the use of a three-dimensional solution. At the same time, however, it is not highly related to the other constructs. Rather, the ideological center of gravity tends to be associated with both dimensions in a similar manner; the correlations are −.48 and −.47, respectively, between the party systems' positions on the center of gravity vector and their positions on Dimensions I and II. The location of the center of gravity vector in Figure 2 reflects this pattern in that it divides the second and third quadrants into relatively equal parts.

If we turn our attention to the clustering of party systems, we gain further insights into the empirical structure of party systems worldwide.[18] Two dominant clusters are evident in Figure 3. One cluster is represented by the more left-oriented nations such as France, West Germany, and Japan, while the second major cluster is represented by the more right-oriented systems such as Ireland, the United States, and Trinidad. In fact, a line drawn perpendicular to the vector representing the ideological center of gravity would perfectly divide the space into the two dominant clusters. It is the ideological center of gravity, therefore, which tends to act as a preliminary structuring principle. The fact that ideological polarization discriminates among party systems primarily in terms of a dichotomy that is highly related to Dimensions I and II explains why only two dimensions are needed to represent the data. Once the party systems are structured in terms of being either

Figure 3 Two-Dimensional Clustering of Forty-Six National Party Systems



less than or greater than the mean of the ideological center of gravity, other structuring principles come into play.[19]

If we examine the left-oriented cluster, we see that almost every party system in it is confined to the second quadrant of the space. That is, party systems with a leftist center of gravity tend to be more highly fractionalized and to exhibit greater ideological polarization. In contrast, party systems with a more rightist center of gravity tend to be spread throughout quadrants one, three, and four. While the more rightist party systems tend to be less fractionalized with less ideological polarization, they are also more variable on each of the dimensions than are party systems in the left-oriented cluster.

Within each of the two dominant clusters, fractionalization and ideological polarization are the bases upon which party systems are differentiated. Within the major rightist cluster three subgroups are found. Party systems in such nations as Austria, Costa Rica, and Colombia have fractionalization scores that are clustered around the mean of all party systems, with ideological polarization scores that tend to be just below the mean of all party systems. Party systems such as those of the United States and Trinidad tend to have the lowest ideological polarization scores, and their fractionalization scores are also quite low. The third group of right-oriented party systems, represented by such nations as Gambia and El Salvador, has the lowest levels of fractionalization and average levels of ideological polarization.

Within the major left-oriented cluster, three subgroups can also be discerned. Members of the first group, represented by systems such as Luxembourg and Switzerland, tend to have the highest levels of fractionalization among the forty-six nations considered here, along with average ideological polarization scores. A second group of systems, including Italy, Malta, and Mauritius, has the highest levels of ideological polarization but only slightly above average levels of fractionalization. The final group of systems, represented by systems such as Greece, Spain, West Germany, and Japan, has above average levels of ideological polarization with fractionalization levels just about average.

In sum, these data indicate that at least three of our four original constructs are important in comparing party systems worldwide. Only party fractionalization and ideological fractionalization are so highly interrelated that it becomes superfluous to use both in classifying party systems. The ideological center of gravity seems to be a particularly important construct with which to begin an analysis of party systems. What appears to be unique about this construct is that it structures party systems worldwide in a dichotomous manner by confining the party systems to a particular region of the space. Systems more left-oriented than average tend to be confined to the area in and around the second quadrant. Systems more right-oriented than average are excluded from the second quadrant. Ideological polarization and party fractionalization are the most important dimensions for discriminating among party systems with the "regions" established by the ideological polarization construct. Within each such region, therefore, one can either directly compare party systems in terms of these constructs or use the constructs to establish subgroups of leftist and rightist party systems.

Conclusion

The purpose of this paper has been essentially conceptual and methodological. Working from a spatial perspective on party systems, in which the key dimensions are articulation and representation, we have developed a set of four constructs for capturing intersystem differences on the two focal dimensions. We have gone on to illustrate how this schema can be used in comparative analysis by applying it to forty-six national party systems. Ours is not the first multidimensional treatment of party systems. However, we think we can fairly claim that it is the first to be based on an overarching conceptualization of party systems in terms of their central components, here defined as representation and articulation. This distinguishes our work from some other recent multidimensional analyses, for example by Powell, who classifies party systems in terms of whether they produce majorities and manifest strong linkages between parties and social groups, and by Flanigan and Zingale, who focus on the social bases of support for parties within a system.[20]

Our data analysis, though intended to be largely illustrative, establishes some empirical points that bear restating. In the first place, it seems that, as of the late 1970s at least, one does not really need the full set of four constructs to differentiate among party systems in various parts of the world. That is, our mental picture of party systems is to some extent more complex than is the "real world" of party systems. This point bears elaboration.

We have proposed four separate measures of party system characteristics: party fractionalization, ideological fractionalization, ideological center of gravity, and ideological polarization. The first of these measures is hardly unique to this paper, and the fourth is akin to a measure proposed by Sigelman and Yough.[21] The second and third measures have not to our knowledge appeared in previous analyses of party systems. Although these characteristics are analytically separable from one another, our empirical analysis of forty-six systems has revealed that there is essentially no difference between party fractionalization and ideological fractionalization. Of course, this finding is subject to change, but it surely does pertain to the party systems we have observed. It also tends to support Downs' speculations concerning the relationship between ideological distributions and the number of parties in a system.[22]

We have also seen that fractionalization of either sort is empirically unrelated to ideological polarization. And we have seen that both fractionalization and ideological polarization are linked to the ideological center of gravity: leftist-oriented systems tend to be more fractionalized and more polarized than do rightist-oriented systems. More generally, our data analysis indicates that it is useful, as an initial gross distinction, to think of party systems in terms of their ideological center of gravity. Once this initial distinction has been made, finer distinctions come into play, relating to the extent of fractionalization of parties and ideologies within the system and the range of ideological-programmatic diversity the system encompasses.

Finally, although the exact placement of various party systems viewed here is in part a function of our use of data from 1978 rather than some other year, the country placements themselves are of interest. Most major western European party systems are part of the left-oriented cluster, but within that cluster we find considerable diversity—perhaps more diversity than is commonly recognized. The customary treatment of these systems is simply to lump them together under the rubric of "multiparty" or "highly fractionalized" systems. These systems clearly *are* highly fractionalized, but this point of similarity should not be allowed to obscure their important points of difference, which emerge quite clearly from our multidimensional analysis. The same point can also be made, with even greater force, about the systems in the right-oriented cluster.

We foresee several different avenues of expansion for the multidimensional perspective illustrated here. First, it can easily be accommodated to longitudinal as well as purely cross-sectional comparison. It is clear that our illustration is time-dependent. If our data base were another year it is likely that the spatial positions of particular party systems would change. The nature of the dimensional structure underlying the space could also possibly need modification. This limitation on our particular illustration, however, points out a distinct advantage of the procedure we have outlined. If one wanted to trace the development of party systems over time,[23] it would be a relatively simple matter to track their "movement" through multidimensional space. Nor need such analyses be wholly descriptive. One might, for example, test theories of party system realignment and dealignment by means of careful examination of movement through multidimensional space over time. One might also—whether longitudinally or cross-sectionally—try to trace the antecedents of particular types of multidimensional party system configurations, asking in effect what demographic, cultural, economic, social, or historical attributes nations share whose party systems resemble one another. Or one might move in the other direction, taking party system configurations as the point of departure for a comparative analysis of policy outputs or system stability. Of course, some starts have already been made in these directions,[24] but it may be appropriate at this point to spell out an illustrative pair of propositions that our analysis suggests.

One proposition, which has its roots in the Sigelman-Yough analysis, is that the extent of ideological-programmatic diversity characteristic of a party system reflects the regime's willingness to tolerate such diversity more than it does the underlying social and economic attributes of the nation. A secondary effect of regime tolerance in the multiparty systems studied here may be on the ideological center of gravity, with more tolerant regimes giving freer play to more leftist-oriented parties. There is less reason, however, to anticipate any close connection between regime tolerance and either party or ideological fractionalization as such; as long as such diversity remains within "acceptable" bounds, even a highly repressive regime may be willing to tolerate it.

A second proposition focuses on the likely connection between the ideological center of gravity and policy outputs. Simply stated, more leftist-oriented systems, as indicated by the ideologies advocated by their parties, should be expected to produce a broader array of social welfare policies and programs. Here again, this is the type of relationship that should hold for this one particular party system attribute but not for the remaining attributes. Many prior cross-national analyses of policy outputs have come up with rather disappointing results concerning the impact of party systems, but the party system characteristics studied in those analyses have usually pertained solely to the fractionalization dimension. If the interpretation offered here is valid, then party system characteristics *do* affect policy outputs, but not *all* party system characteristics have such an effect. Focusing on the ideological center of gravity as the key party system characteristic in this regard may well have a salutary impact on our understanding of national differences in policy outputs.

Again, these propositions are offered solely for illustrative purposes. It is well beyond the scope of this paper to spell out a theory of the causes and consequences of the array of party system characteristics considered here. Ours has been a preliminary mapping operation designed to highlight the primary characteristics of these systems, an operation that seems to us to be a prerequisite of the more theoretically oriented uses to which a multidimensional treatment of party systems can be put.

NOTES

1. For example, Berrigan classified Canadian provinces as "one-party" if the largest party received at least 67 percent of the vote, "modified one-party" if the largest party gained between 62 percent and 67 percent, and so on through "three-party" if the largest party received between 32 percent and 37 percent; see A. John Berrigan, "Interparty Electoral Competition, Stability, and Change: Two-Dimensional and Three-Dimensional Indices," *Comparative Political Studies*, 5 (July 1982), 198–210. However, he made no attempt to explain or defend why he happened to choose these particular cut-off points, which remains a mystery to this day, as does why he selected entirely different (though no less arbitrary) cut-off points in categorizing the states of Australia. More recently, Sartori's long awaited typology of party systems has proven even more problematic. See Giovanni Sartori, *Parties and Party Systems: A Framework for Analysis* (Cambridge: Cambridge University Press, 1976). Not only does he go through intellectual contortions in deciding which parties to include in classifying each system; he also mixes an ideological polarization criterion into some categories but not others and ends up providing no operational rules whatsoever for fitting actual party systems into the categories.

2. For useful reviews see David N. Milder, "Definitions and Measures of the Degree of Macro-Level Party Competition in Multiparty Systems," *Comparative Political Studies*, 6 (January 1974), 431–456; Raimo Valrynen, "Analysis of Party Systems by Concentration, Fractionalization, and Entropy Measures," *Scandinavian Political Studies*, 7 (1972), 137–155; Loren K. Waldman, "Measures of Party Systems' Properties: The Number and Sizes of Parties," *Political Methodology*, 3 (1976), 199–214.

3. One could, of course, use percentage of votes rather than of seats as the basis for analysis, and the choice will have some bearing on the results since the two sets of percentages are not perfectly correlated; see Edward Tufte, "The Relationship between Seats and Votes in Two-Party Systems," *American Political Science Review*, 67 (June 1973), 540–554. Whether seats or votes provide the more useful basis of comparison cannot be decided in isolation from the theoretical content in which one is operating.

4. See especially Kenneth Janda, *Political Parties: A Cross-National Survey* (New York: Free Press, 1980).

5. Anthony Downs, *An Economic Theory of Democracy* (New York: Harper & Row, 1957).

6. Systems omitted for this reason were India, Indonesia, Madagascar, Nauru, Peru, and San Marino.

7. Douglas Rae, "A Note on the Fractionalization of Some European Party Systems," *Comparative Political Studies*, 1 (October 1968), 413–418.

8. However, the value of the party fractionalization index does put a limit on the values that can be assumed by the ideological fractionalization index. If, for example, one party completely dominates the system, then by definition the system will be perfectly nonfractionalized in an ideological sense as well. More generally, for any given political system the ideological fractionalization must always be less than or equal to its party fractionalization.

9. Downs.

10. The problem with including single-party systems in the analysis is that their scores on three of the dimensions are fixed by definition: party fractionalization, ideological fractionalization, and ideological polarization all equal 0. This would make it relatively simple to derive a neat dimensional solution, but that solution would not tell us a great deal.

11. The *Britannica* scores were used in preference to others that have appeared elsewhere because they are available for a large set of nations for a recent year; Lawrence Dodd, *Coalitions in Parliamentary Government* (Princeton: Princeton University Press, 1976); Janda; Lee Sigelman and Syng Nam Yough, "Left-Right Polarization in National Party Systems: A Cross-National Analysis," *Comparative Political Studies*, 11 (October 1978), 355–380.

12. J. B. Kruskal and Frank H. Carmone, *How to Use MDSCAL (Version 5M) and Other Useful Information* (Murray Hill: Bell Laboratories, 1969).

13. The actual stress value can be affected by a number of factors, including the number of points being scaled. According to the guidelines suggested by Kruskal and Carmone, when one scales ten to thirty points in two to five dimensions a stress value of .05 can be considered excellent and a stress value of .10 can be considered good. Since we have a stress value of .09 with our forty-six point, two-dimensional solution, the solution can be considered to be quite good to excellent. For a discussion of stress values and the difference between stress formula one and stress formula two, see Kruskal and Carmone; George B. Rabinowitz, "An Introduction to Nonmetric Scaling," *American Journal of Political Science*, 19 (May 1975), 343–390.

14. For a discussion of varimax rotation and its advantages see Harry H. Harman, *Modern Factor Analysis* (Chicago: University of Chicago Press, 1970). An orthogonal rotation, rather than an oblique rotation, was used because the theoretical advantages of orthogonal dimensions seemed very persuasive in this application. First, the fundamental purpose of most scaling applications is to search for fundamental constructs that provide the basis for a simple structure underlying a complex set of data. Since fundamental constructs are generally viewed as theoretically independent of one another, only orthogonal solutions satisfy this criterion. In this context, it is important to point out that while orthogonal solutions satisfy the theoretical requirement of independent constructs, they do allow dimensions to be correlated in the empirical realm, as is the case in this application. Second, Kaiser points out that the varimax solution satisfies the principle of factorial invariance; Henry F. Kaiser, "The Varimax Criterion for Analytic Rotation in Factor Analysis," *Psychology*, 23 (1958), 187–200. That is, the solution is invariant when the same common factors underlie two different sets of data. Oblique solutions do not satisfy this criterion. Third, the distances among points remain the same under orthogonal rotations; distances among points change with oblique rotations. Finally, Harmon states (p. 273) that "simplicity of interpretation may be offset, however, if the linear description of the variables in terms of correlated factors can be made simpler than in the case of uncorrelated ones." In this particular analysis, we are able to obtain a two-dimensional solution without relinquishing the standard of orthogonality; using correlated factors does not simplify the solution. Thus, there is no readily apparent reason to sacrifice the conceptual simplicity given by an orthogonal solution.

15. The vectors were placed into the two-dimensional space on the basis of a procedure recommended by Rabinowitz.

16. The correlation between ideological and party fractionalization is .83. The strong positive relationship between ideological and party fractionalization is not overly surprising. Numerous theoretical writings, particularly the work of Downs, suggest that there should be a positive relationship between these two constructs. However, we feel that it is important to include both constructs in our analysis. Since the two constructs are not theoretically equivalent, we feel that the distinction must be maintained in any theoretical analysis of party systems. Furthermore, while the two constructs should be related to each other, there are theoretical reasons to suggest why they should not be related to each other in a perfect one-to-one relationship. Factors other than ideology (religion, region, race, tribal custom, and individual personalities) can form the basis for partisan divisions. Even Downs' analysis indicates that one can have a partisan split (two parties) without a fundamental ideological split in either the parties or the electorate. As such, it appears to us that the actual degree of correspondence between party and ideological fractionalization must be examined empirically.

17. Downs.

18. The clustering routine available in the Biomedical Computer Packages was used to undertake the clustering analysis. The average intercluster difference was the option chosen for the analysis.

19. The importance of ideological polarization appears to be that it divides party systems into one of two types. In fact, if one dichotomizes the party systems in terms of those nations above the average on ideological polarization versus those below the average, R^2 between Dimensions I and II and this dichotomy is higher than R^2 between Dimensions I and II and the ideological polarization score treated as a continuum. The correlations between the dichotomy and Dimensions I and II are .45 and .65 respectively. Because of this property, one does not achieve a significant improvement in fit when one uses a three-dimensional solution (stress = .04) instead of a two-dimensional solution (stress = .09). As such, the importance of ideological polarization is not as a dimensional continuum but as a dichotomous phenomenon that tends to "confine" party systems to one of two regions in the space, with discrimination within each region being based upon Dimensions I and II.

20. G. Bingham Powell, "Party Systems and Political System Performance: Participation, Stability and Violence in Contemporary Democracies," *American Political Science Review*, 75 (December 1981), 861–879; William Flanigan and Nancy Zingale, "Western Europe and Anglo-American Party Systems: A Dimensional Analysis," *Comparative Political Studies*, 14 (January 1981), 481–516.

21. Sigelman and Yough.

22. Downs.

23. Shankar Bose, "Political Profiles: An Index for Measuring Similarity over Time," *Comparative Political Studies*, 12 (January 1980), 404–411.

24. Sigelman and Yough; Powell.